

Impose a controlled biaxial strain field on the borosilicate glass liner during high-temperature annealing (e.g., via differential cooling or external mechanical loading) to refine grain size and strengthen grain boundaries, thereby suppressing grain-boundary sliding

Executive Technology Brief

Domain: thermal

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'grain-boundary sliding' concerning borosilicate glass reactor liner. User-submitted engineering problem (session input): "Why do grain-boundary sliding failures limit the service temperature of borosilicate glass reactor liners, and how can the grain structure be refined to fix this?". [cod_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2010-2025). Illustrative records: 'Motion of grain boundaries incorporating dislocation structure'; 'Energy of low angle grain boundaries based on continuum dislocation structure'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Integrated advanced strain engineering combined with thermal annealing to modify grain boundary characteristics

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: Impose a controlled biaxial strain field on the borosilicate glass liner during high-temperature annealing (e.g., via differential cooling or external mechanical loading) to refine grain size and strengthen grain boundaries, thereby suppressing grain-boundary sliding

Governing Model: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: APPLICABLE; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: dT .

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected). Engaged variables: k , dT/dx .

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.